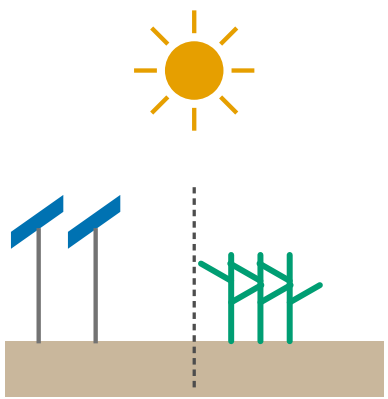
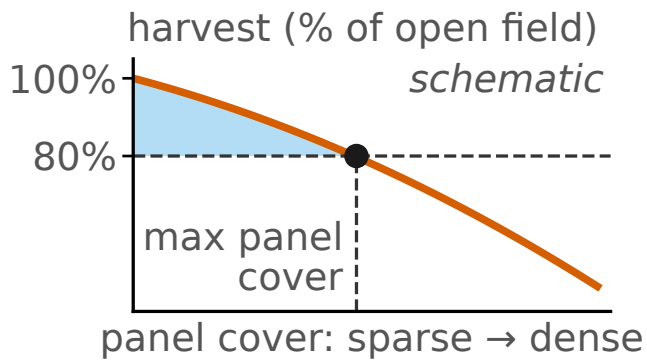
## ❶ How much solar fits on India's farmland and keeps 80% of the harvest?



## ❷ Simulate shade → harvest

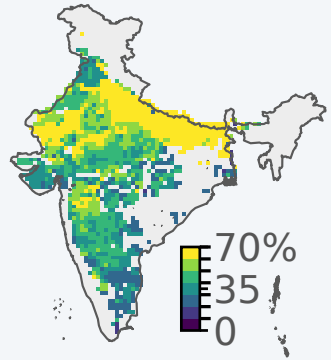



every 0.25° cell · 5 climate models

## ❸ Answer for India

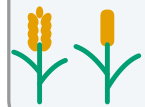
max panel cover keeping 80% harvest



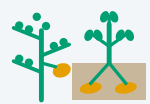

**45.5 TWp**

would generate ≈ 48× India's yearly electricity use **(upper bound)**
2050 (SSP5-8.5): −1% vs 1995–2014

max panel cover:



**~61%** cereals



**~29%** legumes